

Problem

Does there exist a bijection $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that for all distinct A, B ,

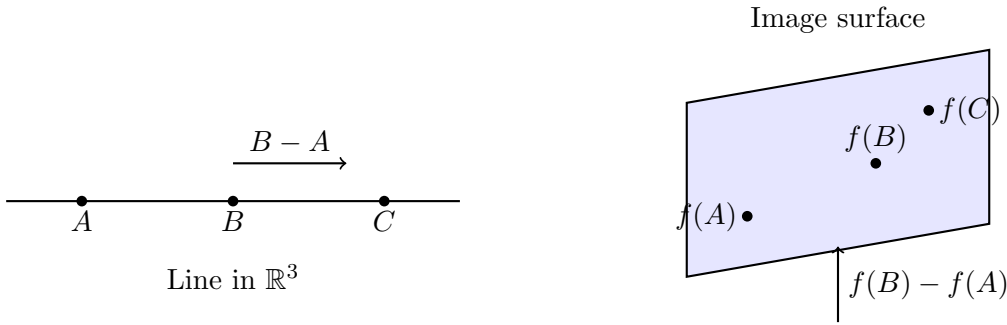
$$\langle B - A, f(B) - f(A) \rangle = 0?$$

Geometric Meaning

For any two points A, B , the condition

$$\langle B - A, f(B) - f(A) \rangle = 0$$

means the displacement vector in the domain is always perpendicular to the displacement vector in the image.



Step 1: Directional Invariance

Take two points A, B that differ only in the i -th coordinate, so $B - A = (b_i - a_i)e_i \parallel e_i$. The perpendicularity condition gives

$$\langle e_i, f(B) - f(A) \rangle = 0 \implies f(B)_i = f(A)_i.$$

That is, the i -th coordinate of $f(x)$ *does not change* when only x_i is varied: $f(x)_i$ is independent of x_i .

Step 2: Contradiction

Take any two distinct points $A \neq B$ differing only in coordinate j , so $a_i = b_i$ for $i \neq j$ and $a_j \neq b_j$.

- For $i \neq j$: since $a_i = b_i$, we have $f(A)_i = f(B)_i$ trivially.
- For $i = j$: by Step 1, $f(x)_j$ is independent of x_j , so varying x_j alone cannot change $f(\cdot)_j$. Hence $f(A)_j = f(B)_j$.

Therefore $f(A) = f(B)$ with $A \neq B$, contradicting the injectivity of f .

Step 3: Substitute into Condition

Let $v = B - A$. Then

$$f(B) - f(A) = A(B - A) = Av.$$

So the condition becomes

$$\langle v, Av \rangle = 0 \quad \forall v \in \mathbb{R}^3.$$

This is equivalent to

$$A + A^T = 0,$$

so A is skew-symmetric.

Step 4: Final Contradiction

A real 3×3 skew-symmetric matrix has determinant 0, hence is not invertible.

Therefore $f(x) = Ax + b$ cannot be bijective.

Answer

No such bijection exists.